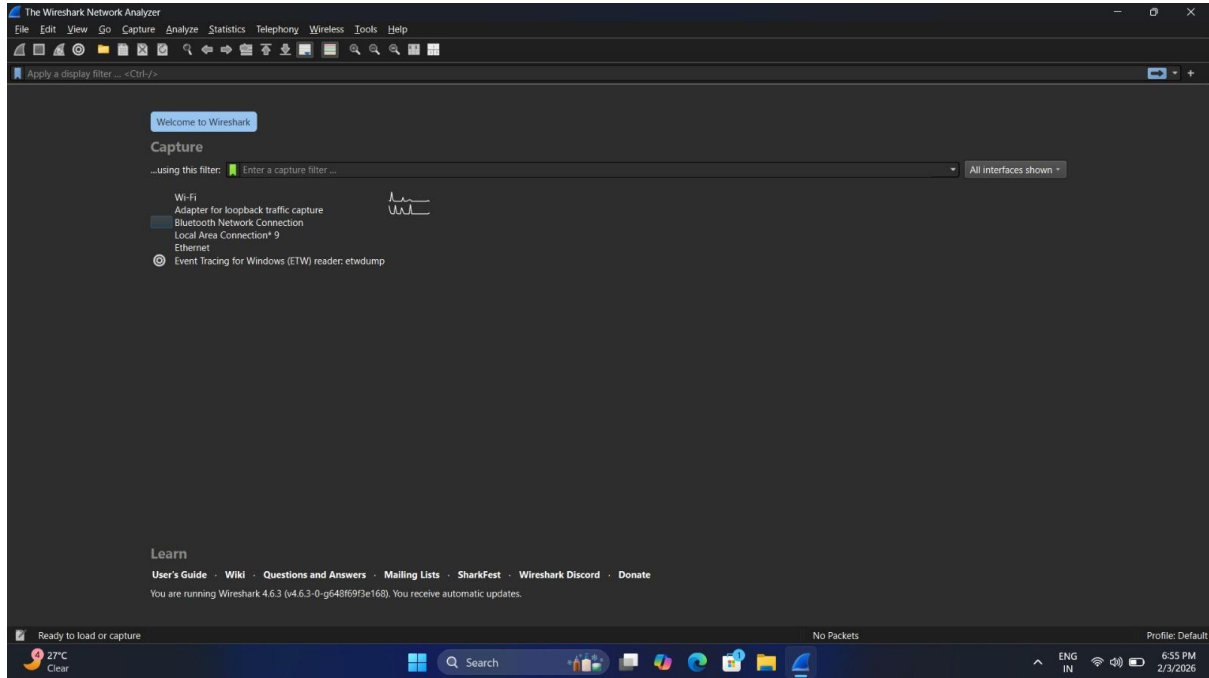


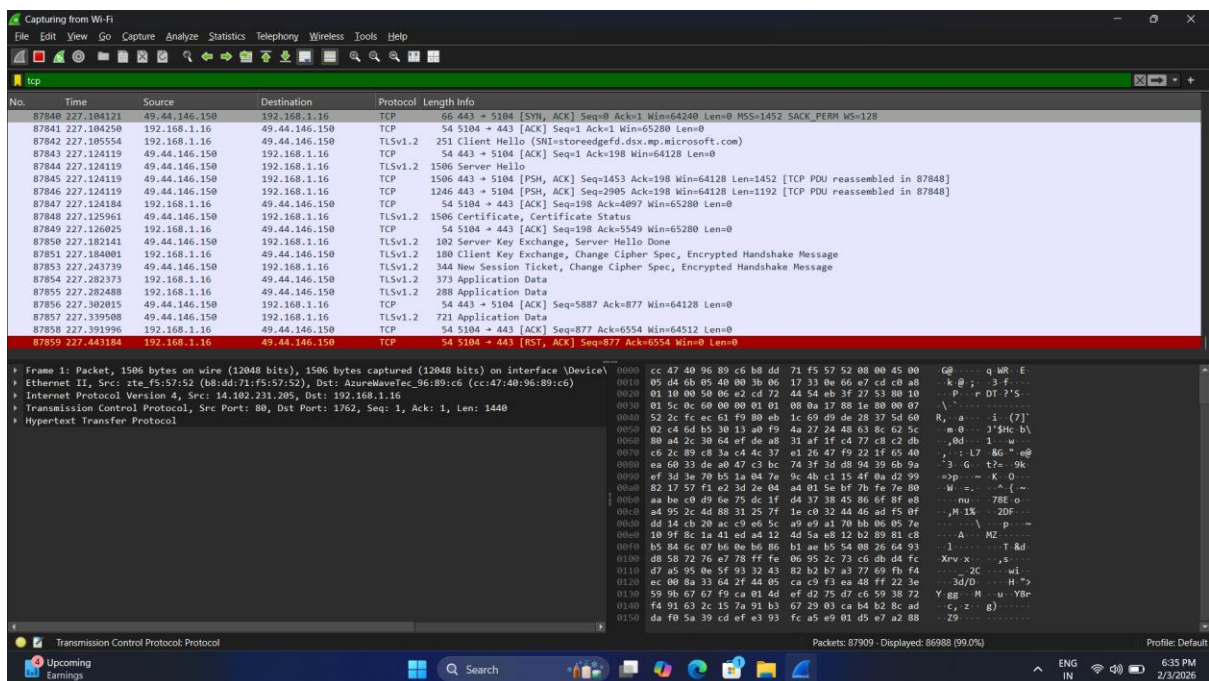
Practical 10

Aim : To capture and analyze network packets using Wireshark and study TCP/TLS and HTTP communication over a Wi-Fi network.

Step 1: Open Wireshark and select the Wi-Fi interface to prepare for packet capture.



Step 2 : Start capturing packets and observe live TCP/TLS traffic between the local system and external servers



Step 3: Apply an HTTP filter to analyze specific HTTP/OCSP requests and responses in detail.

The image shows a Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The main window is divided into three panes: a packet list, a packet details pane, and a packet bytes pane.

The packet list pane shows a list of captured packets. The first packet is selected, and its details are shown in the packet details pane. The packet details pane shows the following information:

- Frame 1301: Packet, 137 bytes on wire (1096 bits), 137 bytes captured (1096 bits) on interface \Device\NPF...
- Ethernet II, Src: AzureWavelec_96:89:c6 (cc:47:40:96:89:c6), Dst: zte_f5:57:52 (b8:dd:71:f5:57:52)
- Internet Protocol Version 4, Src: 192.168.1.16, Dst: 23.38.59.250
- Transmission Control Protocol, Src Port: 5816, Dst Port: 80, Seq: 219, Ack: 1, Len: 83
- [2 Reassembled TCP Segments (301 bytes): #1300(218), #1301(83)]
- Hypertext Transfer Protocol
- Online Certificate Status Protocol

The packet bytes pane shows the raw data of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, Transmission Control Protocol header, and the Hypertext Transfer Protocol body. The packet bytes pane also shows the packet's length (137 bytes) and the reassembled TCP segment (301 bytes).